

Unit 11 - Operations with Rational Numbers

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The following B.E.S.T Standards will be covered in this unit:

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

- *Clarification 1: Instruction includes using one or more operations to solve problems.*

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.

- *Clarification 1: Multi-step expressions are limited to 6 or fewer steps.*

Unit 11, Lesson 1: Adding Rational Numbers



Benchmark

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.



Learning Target

- ☐ I can fluently add rational numbers.



11.1.1 Warm-Up: Landscape Architect

Janice Nicol, a landscape architect, works to design natural systems and outdoor spaces, like trails and parks, for campuses and recreational facilities. Her job is both creative and technical. Creativity is expressed through drawing sketches both by hand and by using computer design programs and imaging. The technical side of her job involves knowing things such as land grading, water calculations, and natural systems.

Janice mentions that she regularly collaborates and works together with others while doing her job.

1. Why do you think it is important to be able to collaborate when doing a job?
2. How can you build your collaboration skills?



11.1.2 Guided Instruction: Adding Positive Rational Numbers

1. Adrian added 5.25 and 6.4. His work is shown.

Adrian's Work	
5.25	
+ 6.4	
5.89	

- a. Write an explanation to Adrian detailing any mistakes he made and how he can correct his work.

- b. What is the sum of 5.25 and 6.4?



When finding the sum of decimal numbers, add values with the same place value.

2. Johannes added $\frac{1}{8}$ and $\frac{3}{4}$. Part of her work is shown.

- a. Complete Johannes' work by filling in the boxes.

Johannes' Answer	
$\frac{1}{8} + \frac{3}{4} =$	
$\frac{1}{8} + \frac{3}{4} \left(\frac{2}{2} \right) =$	
$\frac{1}{8} + \frac{\boxed{}}{\boxed{}} = \frac{\boxed{}}{\boxed{}}$	

- b. Complete the statements by choosing a word or number from the bank. Not all words and numbers may be used, and some may be used more than once.

Word Bank

2 4 8
numerators common denominator
one half fourths seven eighths

To find the sum of $\frac{1}{8}$ and $\frac{3}{4}$, Johannes first rewrote each fraction using a _____.

The _____ of 8 and 4 that Johannes used was _____. Then, she added the two _____ together to find the total number of _____.



11.1.3 Try It: Adding Positive Rational Numbers

- Match pairs of expressions that have the same sum.
Write the equivalent expressions on the lines provided.

$\frac{3}{4} + \frac{4}{8}$	$\frac{2}{3} + \frac{3}{5}$	$\frac{2}{3} + \frac{7}{18}$
$\frac{5}{9} + \frac{1}{2}$	$\frac{15}{15} + \frac{8}{30}$	$\frac{12}{16} + \frac{1}{2}$

_____ = _____

_____ = _____

_____ = _____



11.1.4 Refresher: Operations with Integers

- Work with your partner to complete the table. The first addition expression has been done for you.

Expression	Absolute Value of First Addend	Absolute Value of Second Addend	Addend With Greater Absolute Value	Sign of Sum	Sum
$(-10) + 6$	10	6	-10	-	-4
$10 + (-6)$					
$(-10) + (-6)$					
$10 + 6$					

- How did the size of the absolute value for each addend determine the sign of the sum?
- Explain how finding the sum for the first two expressions was different from finding the sum for the last two expressions.

4. How were the sums for $10 + (-6)$ and $(-10) + 6$ similar?

5. How were the sums for $(-10) + (-6)$ and $10 + 6$ different?

6. Complete the statements using the words *same* or *different* to describe the rules for finding the sums of signed numbers.

When two addends have the _____ sign, the sum will have the sign of the addends.

When two addends have _____ signs, the sum will have the sign of the addend with the greater absolute value.

7. Apply the rules for integers to these expressions with rational numbers. Find the sum of each expression. Then, verify your answer using a calculator.

Addition Expression	Sum When Using the Rule	Sum With a Calculator	Does the Rule Work?
$(-0.92) + 50.6$			
$4.95 + 3.05$			
$\left(-\frac{4}{5}\right) + \left(\frac{22}{25}\right)$			
$1.25 + (-6.45)$			
$\left(-\frac{2}{3}\right) + \left(-\frac{9}{20}\right)$			
$(-17.01) + (-2.99)$			

8. Complete the sentence.

The sign rules for adding integers are

- ☐ the same
- ☐ not the same

as the rules for adding rational numbers.



11.1.5 Guided Practice: Applying Signed Number Rules to Rational Numbers

1. The blanks in the addition equation represent missing values. Fill in the blanks for the addition equation below to make a true statement.

$$-12\frac{3}{4} + \left(-5\frac{\boxed{}}{6}\right) = -18\frac{\boxed{}}{12}$$

2. Find the sum for each addition expression.

Addition Expression	Sum
$\frac{4}{5} + \left(-\frac{3}{7}\right)$	
$-9\frac{1}{2} + \left(-14\frac{1}{8}\right)$	

3. The blanks in the addition equation represent digits in the values.

- a. Fill in the blanks for the addition equation below to create a true equation.

$$-16.537 + 6.1\boxed{}8 = -10.\boxed{}59$$

- b. Describe the process you used to determine the values for each of the blanks above.

- c. Discuss your process with your partner. Write a summary of their process. Have your partner initial to show that your summary accurately expressed their process.

Your Partner's Process

Partner's
Initials _____



11.1.6 Your Turn!

1. Find the sum for each expression.

a. $-1\frac{1}{2} + (-\frac{1}{8})$

b. $-12.37 + 18.161$

c. $-\frac{2}{5} + (-7\frac{1}{4})$

d. $-4.48 + 9.366$

e. $-233.24 + 25.8$

f. $-4\frac{1}{7} + 17\frac{2}{9}$



11.1.7 Wrap-Up: Ticket Out the Door

1. Choose a value from Row A and a value from Row B that, when added, will result in a negative answer. Then, calculate the sum.

Row A

$-1\frac{3}{5}$	$-\frac{1}{8}$	$-2\frac{1}{4}$
-----------------	----------------	-----------------

Row B

$1\frac{1}{2}$	$2\frac{6}{25}$	$\frac{19}{40}$
----------------	-----------------	-----------------

Row A

Row B

Sum

+

=

Unit 11, Lesson 2: Subtracting Rational Numbers



Benchmark

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.



Learning Target

☐ I can fluently subtract rational numbers.



11.2.1 Warm-Up: Math Talk

Several students discussed which fraction was greater, $\frac{13}{19}$ or $\frac{2}{3}$. A summary of the discussion is shown.

Amy:	I changed $\frac{13}{19}$ to $\frac{13}{18}$, and since $\frac{2}{3}$ is equal to $\frac{12}{18}$, I'm about 70% confident that $\frac{13}{19}$ is greater than $\frac{2}{3}$.
Brandon:	The fractions were small enough that I could do mental math and change them into common denominators, so $\frac{13}{19} = \frac{39}{57}$, and $\frac{2}{3} = \frac{38}{57}$. Therefore, $\frac{13}{19}$ is greater than $\frac{2}{3}$.
Carlos:	I changed $\frac{13}{19}$ to $\frac{14}{20}$. Then I thought of a pizza, and that's a lot of little pieces. So, I'm about 70% confident that $\frac{2}{3}$ is bigger than $\frac{13}{19}$.
Dimitri:	I know $\frac{2}{3}$ is $\frac{12}{18}$ if I multiply the top and bottom by 6. $\frac{12}{18}$ is bigger than $\frac{13}{19}$. I'm 90% confident.

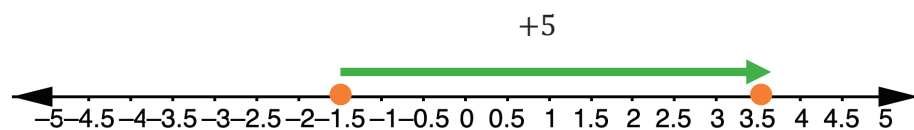
1. Explain which student's reasoning you agree with.
2. Choose one student you didn't agree with, and explain why you did not agree.



11.2.2 Guided Instruction: Understanding Differences

Finding the **difference** of two numbers means to *subtract the numbers*. When finding the difference between two numbers, subtract the numbers in the order they are named. For example, the difference of 3.5 and -1.5 is represented by the subtraction expression $3.5 - (-1.5)$.

The difference of two numbers is how far apart the numbers are on a number line.



1. The numbers 3.5 and -1.5 are _____ units apart. The direction to move from the second number (-1.5) to the first number (3.5) is to the ☐ left, ☐ right, or the ☐ positive ☐ negative direction. Therefore $3.5 - (-1.5) = +5$.

2. Suppose the numbers are subtracted in the opposite order, $(-1.5) - 3.5$.

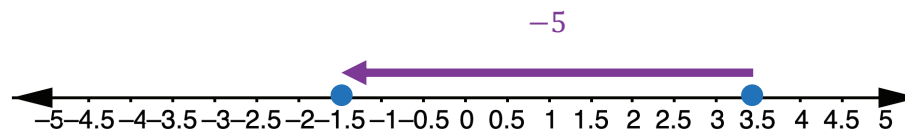
The numbers 3.5 and -1.5 are still _____ units apart, but

the direction of the move has changed. To move from

the second number (3.5) to the first number (-1.5) is to

move to the ☐ left ☐ right or the ☐ positive ☐ negative direction.

Therefore, $(-1.5) - 3.5 = -5$.





11.2.3 Collaboration: Understanding Differences

- Working with your partner, write each phrase as a subtraction expression. Then, draw a number line representation to find the value of the expression. The first phrase has been done for you.

Phrase	Subtraction Expression	Number Line Representation
The difference of $-\frac{1}{5}$ and $-\frac{4}{5}$	$-\frac{1}{5} - (-\frac{4}{5})$	
Value = $\frac{3}{5}$		

Phrase	Subtraction Expression	Number Line Representation
The difference of $-\frac{4}{5}$ and $-\frac{1}{5}$		
Value = _____		

Phrase	Subtraction Expression	Number Line Representation
The difference of 0.9 and -0.5		
Value = _____		

Phrase	Subtraction Expression	Number Line Representation
The difference of -0.5 and 0.9		
Value = _____		

2. With your partner, complete the table to find the difference of the values.

Expression	Distance Between the Values	Direction from Second Value to First Value	Difference
$9.9 - 6.2$	3.7	Right/positive	$9.9 - 6.2 = 3.7$
$9.9 - (-6.2)$			$9.9 - (-6.2) = \underline{\hspace{2cm}}$
$(-9.9) - 6.2$			$(-9.9) - 6.2 = \underline{\hspace{2cm}}$
$(-9.9) - (-6.2)$			$(-9.9) - (-6.2) = \underline{\hspace{2cm}}$
$6.2 - 9.9$			$6.2 - 9.9 = \underline{\hspace{2cm}}$
$6.2 - (-9.9)$			$6.2 - (-9.9) = \underline{\hspace{2cm}}$
$(-6.2) - 9.9$			$(-6.2) - 9.9 = \underline{\hspace{2cm}}$
$(-6.2) - (-9.9)$			$(-6.2) - (-9.9) = \underline{\hspace{2cm}}$



11.2.4 Guided Instruction: Finding Differences

Adding a negative value has the same result as subtracting a positive value. This is because addition and subtraction have an inverse relationship.

For example, $5 + (-6)$ is equivalent to $5 - 6$. Both expressions have a value of -1 .

This relationship can be used to subtract rational numbers. The difference of two numbers can be rewritten as an equivalent addition expression. Then, the rules for addition of two numbers can be followed.

1. Fill in the blanks to complete the table.

Original Expression	$1\frac{2}{5} - \left(-\frac{6}{7}\right)$
Rewrite as an equivalent addition problem by keeping the first term, changing subtraction to addition, and changing the second term to its opposite.	$1\frac{2}{5} \square \left(\square \frac{6}{7}\right)$
Solve the problem using the rules of addition.	$\square$



11.2.5 Collaboration: Does the Order Matter?

1. With your partner, determine who will be Partner A and who will be Partner B. Partner A will work the problems in Column A while Partner B works the problems in Column B. Then, work together to answer the questions that follow.

Column A	Value	Column B	Value
$3 - 2$		$2 - 3$	
$5 - (-9)$		$(-9) - 5$	
$(-11) - 2$		$2 - (-11)$	
$(-\frac{1}{6}) - (-\frac{1}{3})$		$(-\frac{1}{3}) - (-\frac{1}{6})$	
$(-1.2) - (-3.6)$		$(-3.6) - (-1.2)$	
$(-2\frac{1}{2}) - (-3\frac{1}{2})$		$(-3\frac{1}{2}) - (-2\frac{1}{2})$	

2. Explain the difference between the expressions in Column A and the expressions in Column B.
3. Explain the relationship between the values of the expressions in Column A and the value of the expressions in Column B.



11.2.6 Your Turn!

1. What is the difference of $\frac{9}{14}$ and $-\frac{13}{20}$?
2. Find the difference for each expression.
 - a. $5\frac{7}{8} - (-2\frac{2}{5})$
 - b. $-45.6 - (-1.772)$



11.2.7 Wrap-Up: Growth Mindset

Making mistakes is one of the best ways to learn. Reflecting on our mistakes helps our brains to look for that mistake again and to avoid it the next time.

1. What mistake did you make today that you can learn from?

Means

I

Start

To

Acquire

Knowledge

Experience

Skills

Unit 11, Lesson 3: Multiplying Rational Numbers



Benchmark

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.



Learning Target

- ☐ I can fluently multiply rational numbers.



11.3.1 Warm-Up: Bell Work

1. Using the digits 1 to 6 at most one time each, fill in the boxes so that the top two equations are equal, and the bottom equation has the greatest value.

$$- \boxed{} + \boxed{} =$$

$$- \boxed{} - \boxed{} =$$

$$- \boxed{} - - \boxed{} =$$



11.3.2 Collaboration: Signed Number Rules

1. Fill in the blank with the words *positive* or *negative* to describe the rules for multiplying integers.
- The product of two positive integers is a _____ integer.
 - The product of two negative integers is a _____ integer.
 - The product of a negative integer and a positive integer is a _____ integer.
2. Discuss with your partner whether you think the same rules apply to rational numbers. Write a summary of your reasoning.

3. Complete the table. Use a calculator to find the product of each expression to verify if the same rules for signs apply when multiplying rational numbers.

Expression	Sign of Product Using the Rules	Product	Did the Same Rule Apply?
$\left(-\frac{3}{5}\right)\left(-\frac{3}{11}\right)$			
$\frac{7}{18}\left(-4\frac{8}{25}\right)$			
$21.285 \cdot ^{-}3.8$			
$(^{-}91.09)(^{-}6.4)$			

4. Complete the sentence.

The rules for multiplying rational numbers are

- ☐ the same as
- ☐ different from

the integer rules for multiplication.



11.3.3 Guided Practice: Applying Signed Number Rules to Rational Numbers

1. With your partner, decide who will be Partner A and who will be Partner B. Then, find the product of the expression in the column that matches your letter.

Column A	Product	Column B
$790\left(\frac{1}{10}\right)$		$7.9(10)$
$\left(-\frac{6}{7}\right) \cdot 7$		$(0.1) \cdot (-60)$
$2.1 \cdot (-2)$		$(-8.4)\left(\frac{1}{2}\right)$
$2.5 \cdot ^{-}3.25$		$\left(-\frac{5}{2}\right)\left(\frac{13}{4}\right)$

The expressions in each row have the same product. Check your work with your partner and, if necessary, resolve any differences. Write the agreed-upon product in the center column.

2. Choose a value from Row A and a value from Row B that, when multiplied, will result in a negative answer. Then, calculate the product.

Row A

$-9\frac{2}{7}$	$-\frac{3}{5}$	$12\frac{2}{9}$
-----------------	----------------	-----------------

Row B

$4\frac{3}{8}$	$\frac{11}{16}$	$-\frac{9}{50}$
----------------	-----------------	-----------------

Row A

×

Row B

=

Product



11.3.4 Wrap-Up: Ticket Out the Door

1. Find the product of each expression.

a. $(0.24)(-3.8)$

b. $\left(-\frac{2}{5}\right)\left(-\frac{9}{8}\right)$

2. Fill in the blanks with the words *positive* or *negative*.

- Two positive factors result in a _____ product.
- Two negative factors result in a _____ product.
- A positive factor and a negative factor result in a _____ product.

Unit 11, Lesson 4: Dividing Rational Numbers



Benchmark

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.



Learning Target

☐ I can fluently divide rational numbers.



11.4.1 Warm-Up: Would You Rather?

1. Explain if you would rather have a stack of quarters that stands from the floor to the top of your head, or \$225.00.
2. Explain whether you would change your mind if the quarters were stacked to the top of your desk.



11.4.2 Refresher: Dividing Positive Rational Numbers

1. Ren found the following quotient, making at least one error in her work.
 - a. Circle the error(s) in Ren's work and write an explanation of the error(s). Then, find the correct quotient.

Ren's Work	Explanation	Correct Quotient
$12.636 \div 3.12$ $ \begin{array}{r} .040 \\ 3.12 \overline{)12.636} \\ \underline{-1248} \\ 156 \\ \underline{-0} \\ 0 \end{array} $		

- b. What advice do you have for Ren that could help her with dividing decimals in the future?

2. Trevor found the following quotient, making at least one error in his work. Circle the error(s) in Trevor's work. Write an explanation of the error(s). Then, find the correct quotient.

Trevor's Work	Explanation	Correct Quotient
$ \begin{aligned} &\left(\frac{38}{13}\right) \div \left(1\frac{6}{13}\right) \\ &\left(\frac{38}{13}\right) \div \left(\frac{19}{13}\right) \\ &\frac{(38 \div 19)}{13} \\ &\frac{2}{13} \end{aligned} $		



11.4.3 Collaboration: Signed Number Rules

1. Fill in the blanks with the words *positive* or *negative* to write the rules for dividing integers.
 - The quotient of two positive integers is a _____ integer.
 - The quotient of two negative integers is a _____ integer.
 - The quotient of a negative integer divided by a positive integer is a _____ integer.
 - The quotient of a positive integer divided by a negative integer is a _____ integer.
2. Discuss with your partner whether you think the same rules apply to rational numbers. Write a summary of your reasoning.

3. Complete the table. Use a calculator to find the quotient of each expression to verify if the same rules for signs apply when dividing rational numbers.

Expression	Sign of Quotient Using the Rules	Quotient	Did the Same Rule Apply?
$\left(-2\frac{3}{8}\right) \div \left(-1\frac{1}{4}\right)$			
$6\frac{7}{18} \div -4\frac{2}{9}$			
$43.008 \div -4.096$			
$(-525.6) \div (-0.36)$			

4. Complete the sentence.

The rules for dividing rational numbers are

- ☐ the same as
- ☐ different from

the integer rules for division.



11.4.4 Guided Practice: Finding the Quotients of Rational Numbers

1. Circle the expressions that will result in a positive quotient.

$$^{-}9.655 \div 0.5$$

$$0.068 \div ^{-}0.01$$

$$^{-}14.628 \div ^{-}4.6$$

$$\frac{16}{20} \div \left(-\frac{8}{15}\right)$$

$$\left(-\frac{9}{6}\right) \div \left(-\frac{5}{3}\right)$$

$$\left(-\frac{3}{5}\right) \div 2\frac{1}{2}$$

- 2.
- a. Write a division sentence in the form of $a \div b$ where the value of a is a number **between** 15.1 and 16.2, and b is a number **between** $^{-}4.6$ and $^{-}3.3$.
- b. What is the value of the quotient?



11.4.5 Your Turn!

1. Find the value of each expression.

a. $\left(-3\frac{5}{9}\right) \div \left(3\frac{1}{3}\right)$

b. $70.62 \div ^{-}160.5$

c. $\left(-1\frac{6}{7}\right) \div \left(-5\frac{3}{4}\right)$

d. $^{-}44.64 \div ^{-}2.88$

e. $8\frac{1}{5} \div \frac{3}{2}$

f. $-45.018 \div (-7.32)$



11.4.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is in relation to each statement. Then, provide a brief explanation or evidence to support your self-assessment.

1. I can correctly decide the sign of the quotient when dividing rational numbers.

I need help.
I can't get started.

I'm getting there.

I understand and have evidence.



Evidence:

2. I can fluently divide rational numbers that are fractions.

I need help.
I can't get started.

I'm getting there.

I understand and have evidence.



Evidence:

3. I can fluently divide rational numbers that are decimals.

I need help.
I can't get started.

I'm getting there.

I understand and have evidence.



Evidence:

Unit 11, Lesson 5: Solving Real-World Problems with Rational Numbers – Part 1



Benchmark

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.



Learning Target

- ☐ I can solve problems with one or two operations by adding, subtracting, multiplying, and/or dividing rational numbers.



11.5.1 Warm-Up: Notice and Wonder

Several division equations are shown.

$$0.28 \div 0.07 = \boxed{}$$

$$2.8 \div 0.7 = \boxed{}$$

$$28 \div 7 = 4$$

$$280 \div 70 = 4$$

$$2,800 \div 700 = 4$$

$$28,000 \div 7,000 = 4$$

1. What do you wonder about the pattern?
2. What do you notice about the equations?
3. Use the pattern in the equations to determine the quotients of the first two equations.

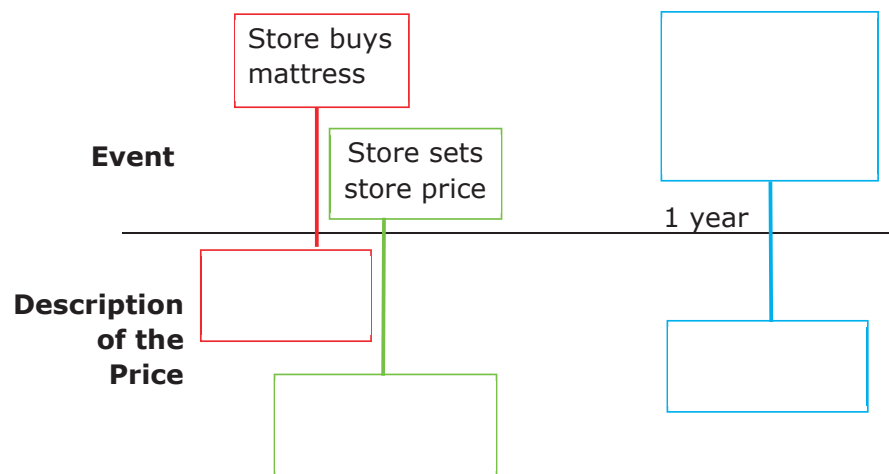


11.5.2 Collaboration: Solving a Problem with Rational Numbers

In order to make a profit, stores sell products for more than what they paid for them. For example, a local furniture store buys a mattress from a manufacturer for a certain price. The store then sets the store price at $2\frac{1}{2}$ times what they paid the manufacturer for it. If the mattress hasn't sold in a year, then the store will reduce its price by $\frac{1}{3}$ of the original store price.

Work with your partner to complete the following based on the information given.

1. Drawing a diagram can help make sense of the problem. Without calculating, complete the timeline to show how the price of a mattress the store purchases for \$325.50 will change over time.



2. Explain whether the store's first price will be greater than or less than the price they paid the manufacturer.

3. What is the first price the store will set for the mattress?

4. Explain whether the price after a year will be greater than or less than the store's first price.

5. What is the sale price of the mattress after one year?

6. If the store sells the mattress after one year, explain if the store will make or lose money on the mattress.



11.5.3 Guided Instruction: Solving Real-World Problems with Rational Numbers

1. Alexis has a monthly subscription to the video streaming service ViewNow for \$7.20 per month. For her birthday she received a ViewNow gift card in the amount of \$55.00.
 - a. Write an expression using addition that represents the balance on the card after one month if the only charge on the gift card is the monthly subscription.
 - b. Complete the table by filling in values on each blank line to record the activity on the account each month. Continue filling in the table to determine how many months it will take before Alexis does not have enough money to pay the monthly fee using the gift card. Part of Month 1 has been completed for you.

Month	Starting Balance	Change in Account	Ending Balance
1	\$55.00	$\$55.00 + \underline{\hspace{2cm}}$	$\$ \underline{\hspace{2cm}}$
2	$\$ \underline{\hspace{2cm}}$	$\$ \underline{\hspace{2cm}} + \underline{\hspace{2cm}}$	$\$ \underline{\hspace{2cm}}$
3	$\$ \underline{\hspace{2cm}}$	$\$ \underline{\hspace{2cm}} + \underline{\hspace{2cm}}$	$\$ \underline{\hspace{2cm}}$

- c. How many months will it take before Alexis does not have enough money left on the gift card to pay the monthly charge?

2. The **departure from the average** is the difference between the actual amount of rain and the average amount of rain for a given month. The departure from the average rainfall, in inches (in.), for Orlando, Florida for June, July, and August is shown in the table.

June	July	August
$-3\frac{19}{20}$ in.	$-1\frac{7}{10}$ in.	$4\frac{5}{8}$ in.

- a. If the average rainfall for June is $7\frac{29}{50}$ in., what was the actual rainfall in June?
- b. What is the difference in the departure from the average rainfall from July to August?

3. Ramon paid \$18,360.00 for a new car. By the end of the first year, the value of the car changed by $-\frac{3}{14}$ of its original value. By the end of the second year, the value of the car changed by $-\frac{2}{9}$ of its value at the end of the first year.

- a. Is the value of the car increasing or decreasing during these two years?
- b. By what amount did the value of the car change at the end of the first year?
- c. What is the value of Ramon's car at the end of the second year?

4. A water well drilling rig dug to a depth of -147.5 feet (ft). The rig drills $-2\frac{1}{2}$ ft in one hour.

a. How many hours has the drill been running?

b. At this rate, how many hours will it take until the drill reaches a depth of -250 ft?



11.5.4 Your Turn!

1. The table shows the change in value for a share of stock in a company at the end of each week during the month of December. A negative value represents the amount the share decreased, and a positive value represents the amount the share increased.

December
$-\$22.65$
$\$225.30$
$-\$106.42$
$\$91.10$

If the value of the stock began at $\$75.37$ at the start of the month, what is the final value of the share at the end of the month?

2. Samuel is practicing for his town's annual high school quarterback football-throwing competition. Last year's champion threw the football $40\frac{1}{4}$ yards (yd). A summary of Samuel's throws in relation to the throws of last year's champion is shown. Complete the table to determine how far Samuel threw the football during each of his tries.

Throw	Comparison of Samuel's Throw to Last Year's Champion	Expression	Yards Thrown
1 st	$+3\frac{1}{3}$ yards	$40\frac{1}{4} + 3\frac{1}{3}$	
2 nd	$-\frac{9}{7}$ yards	$40\frac{1}{4} + \left(-\frac{9}{7}\right)$	
3 rd	$-2\frac{6}{13}$ yards		

3. Delaney wants to sell her bicycle. The original price of the bike was \$150. The value of the bike has changed by $-\frac{2}{5}$ of the original price. What is the current value of the bike?
4. At the beginning of the day on Monday, Stephon had \$15.78 in his lunch account. At the end of the day on Friday, he had \$1.53 in his lunch account. If Stephon purchased the same lunch each day on Monday through Friday, describe how the account changed each day.



11.5.5 Wrap-Up: Growth Mindset

1. As you reflect on today's lesson, circle the word *Sometimes*, *Always*, or *Never* to indicate how often each statement describes you.

When I work hard on a problem, I expect that the answer will be correct.	Sometimes	Always	Never
There is no point in trying to do a problem if I know I won't get the correct answer.	Sometimes	Always	Never
Even if I made a mistake, I can still learn from working through a problem.	Sometimes	Always	Never
I can show that I understand how to solve a problem, even if there is an error in my work.	Sometimes	Always	Never
If my answers are incorrect even after working extremely hard, I still choose to work hard on the next problem.	Sometimes	Always	Never

Unit 11, Lesson 6: Solving Real-World Problems with Rational Numbers – Part 2



Benchmark

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.



Learning Target

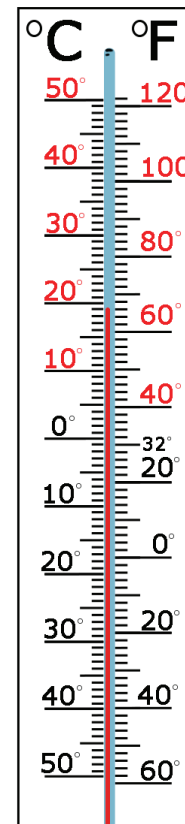
- ☐ I can solve problems with three operations involving any of the four operations with rational numbers.



11.6.1 Warm-Up: Math Fail

A picture of a thermometer is shown.

1. Explain the error in the image.



2. Describe how to fix the image so that there is no “math fail.”



11.6.2 Activity: Error Analysis

With your partner, answer each question below.

1. Diana was given the following problem to solve.

Eugene has \$40.00 to spend at a theme park gift shop. He receives a discount of $\frac{1}{5}$ off the price of an item for being an annual passholder. He wants to purchase a collectible passholder pin that has a retail price of \$19.95. How much will he have left after purchasing the pin?

Diana solved the problem with the steps shown.

Diana's Work	
$19.95 \cdot (-\frac{1}{5})$ $\frac{19.95 \cdot (-\frac{1}{5})}{1}$ $\frac{-19.95}{5}$ $5 \overline{)19.95}$ $\begin{array}{r} 3.99 \\ -15 \\ \hline 49 \\ -45 \\ \hline 45 \\ -45 \\ \hline 0 \end{array}$	$\$40.00 - \3.99 $\begin{array}{r} 399 \\ -399 \\ \hline 3601 \end{array}$ Eugene had \$36.01 left.

- a. Identify what Diana did correctly and incorrectly.

Correct	Incorrect

- b. How much money will Eugene have left over after buying the pin? Show your work.

2. Jeffrey was given the following problem to solve.

Diego is planning to hike the very difficult Shorty's Well Route to Telescope Peak, which begins at an elevation of -262 feet and ends at an elevation of 11,043 feet. He wants to hike $\frac{3}{5}$ of the change in elevation of the trail on the first day of his hike. If Diego sticks to his plan, at what elevation will he start on the second day?

Jeffrey solved the problem with the steps shown.

Jeffrey's Work	
$11043 - 262 =$ $\begin{array}{r} 11043 \\ -262 \\ \hline 10781 \text{ feet} \end{array}$ $\frac{10781}{1} \cdot \frac{3}{5} = \frac{32343}{5}$	$\begin{array}{r} 6468.6 \text{ feet} \\ 5 \overline{)32343.0} \\ -30 \\ \hline 23 \\ -20 \\ \hline 34 \\ -30 \\ \hline 43 \\ -40 \\ \hline 30 \\ -30 \\ \hline 0 \end{array}$ Diego plans to be at 6468.6 feet to start Day 2.

- a. Identify what Jeffrey did correctly and incorrectly.

Correct	Incorrect

- b. At what elevation will Diego start on the second day? Show your work.

3. Ashlyn was given the following problem to solve.

Fernando runs after school to prepare for the cross-country season. On Monday he ran 4.34 miles, Wednesday he ran 6.56 miles, and on Friday he ran 1.375 miles. His teammate Marcus only ran $\frac{4}{10}$ the number of miles as Fernando. How many miles did Marcus run?

Ashlyn solved the problem with the steps shown.

Ashlyn's Work	
$\begin{array}{r} 4.34 \\ +6.56 \\ \hline 10.90 \\ +1.375 \\ \hline 23.65 \text{ miles} \end{array}$	$\begin{array}{l} 23.65 \div \frac{4}{10} \\ 23.65 \cdot \frac{10}{4} = \frac{236.5}{4} \\ \begin{array}{r} 59.0 \\ 4 \overline{)236.5} \\ \underline{-20} \\ 36 \\ \underline{-36} \\ 0 \end{array} \end{array}$ Fernando ran 59 miles.

- a. Identify what Ashlyn did correctly and incorrectly.

Correct	Incorrect

b. How many miles did Marcus run? Show your work.

a. Identify what Craig did correctly and incorrectly.

Correct	Incorrect

4. Craig was given the following problem to solve.

Rhonda makes a specific hair dye color by mixing $1\frac{1}{8}$ ounces (oz) of red tone, 0.75 oz of gray tone, and $\frac{5}{8}$ oz of brown tone. If Rhonda needs to make 20 oz of the same color dye, how much of each color tone is needed?

Craig solved the problem with the steps shown.

Craig's Work	
$\frac{1}{8} + \frac{5}{8}$ $\frac{8}{8} + \frac{5}{8} = \frac{13}{8}$ $\frac{13}{8} = \frac{5}{4}$	$\frac{5}{4} + 0.75$ $1\frac{1}{4} + 0.75$ $1.25 + 0.75$ $\begin{array}{r} 1.25 \\ + 0.75 \\ \hline 2.00 \end{array}$ ounces total
$20 \div 2 = 10 \text{ ounces needed of each}$	$\frac{1}{8} \cdot 10 = 1\frac{2}{8} \text{ oz red}$ $0.75 \cdot 10 = 7.5 \text{ oz gray}$ $\frac{5}{8} \cdot 10 = \frac{50}{8} = 6\frac{2}{8} \text{ oz brown}$

b. How much of each color is needed to make 20 oz of dye? Show your work.





11.6.3 Collaboration: Finding Missing Values

Many homeowners have solar panels that can generate energy to power their homes. The Better Utility Company charges customers who have solar panels using the following criteria:

- Customers are charged \$0.12 per kilowatt-hour for the energy used that is provided by the utility company.
- The utility company does not charge customers for energy they use from their own solar panel(s).
- For every kilowatt-hour of electricity that customers with solar panels generate but do not use, the utility company gives a price adjustment of $-\$0.025$.

Erin Hill's bill for the month of July is missing some information.

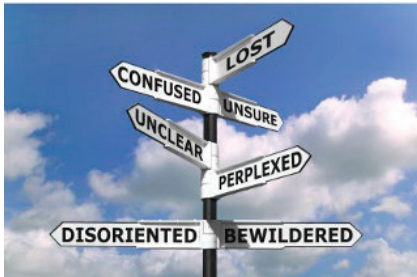
Best Utility Company 123 Ohms Way Big City, ST 12345	Erin Hill 456 Long St. Small Town, ST 98765
Kilowatt-hours used from Best Utility Company _____	Charge \$82.04
Kilowatt-hours generated by solar panel and not used _____ by customer	Credit \$ ⁻ 4.10
	Amount Due \$ _____

1. Work with your partner to fill in the missing information on the bill. Round answers to nearest hundredth.
2. Find another classmate other than your partner and compare your missing values.
 - If the missing values for both are the same, have your classmate initial next to each value in your bill.
 - If the missing values are different, discuss your work to determine what error(s) were made. After correcting any mistakes, initial next to each value in each other's bill.



11.6.4 Wrap-Up: Reflection

1. Complete the table by reflecting on the day's lesson.

Today I deserve  for ...	
Today I  by doing ...	
Today I was  by ...	



Unit 11, Lesson 7: Solving Real-World Problems with Rational Numbers – Part 3



Benchmark

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.



Learning Target

- ☐ I can solve problems with at least three operations involving any of the four operations with rational numbers.



11.7.1 Warm-Up: Notice and Wonder

1. What do you notice about the binder clips in the image?



2. What questions come to mind when you look at this image?
3. The stack of papers contains about 260 sheets of paper. Estimate the number of sheets each binder clip can hold.
4. To make a good estimate, what information would be useful to know?
5. What tools or resources could help you find that information?



11.7.2 Activity: Gallery Walk

Your teacher will assign your group one of the following scenarios. Work together to answer the question(s), writing neatly on the chart paper provided. Show all your work to prepare a presentation for the other students in your class.

Marie plans to use a ride sharing service for the trip from Tampa International Airport to her hotel. The service has a booking fee of \$2.45 and a cost of \$0.42 for the first $\frac{1}{4}$ mile and an additional fee of \$0.16 for each 0.1 mile after the first $\frac{1}{4}$ mile.

1. How much will the trip cost if Marie's hotel is 7.25 miles from the airport?

Scenario 1

2. Marie decides to stop by a grocery store before arriving at her hotel. The ride sharing service charges \$0.45 for each minute they have to wait on a customer. Her total bill was \$21.72. How long was Marie in the grocery store?

**Scenario
2**

Leah and her parents have an agreement. She borrowed money from them to buy a video gaming system and is supposed to pay them back within 3 months. If Leah does not have the balance paid by the agreed-upon time, she will have to give the system to her parents until it is paid off. Leah and her parents created a payment plan to help her meet this goal. Starting on July 1, Leah should pay \$5.95 per day until the gaming system is paid off. On July 5th, her balance was $-\$10.25$ since she couldn't pay the full amount each day. Leah was not able to pay anything again for 15 days. On July 20th, Leah paid \$85.00.

1. How much had Leah paid her parents by July 5th?
2. By July 20th, how much more or less had Leah paid than what was required by the payment plan?

A coffee shop keeps track of changes in their coffee inventory. A week of their record-keeping is shown.

Day	Change in Inventory (lb)
Monday	150
Tuesday	-43.3
Wednesday	-50.5
Thursday	72.8
Friday	-52.7
Saturday	-21.6
Sunday	0

**Scenario
3**

1. How much coffee is left at the end of the week?
2. Coffee is delivered in 50-pound (lb) bags, 3 bags at a time. How many pounds of coffee were sold on Thursday?

Scenario 4

Recall from the previous lesson that The Better Utility Company charges \$0.12 per kilowatt-hour for the energy a customer uses that is provided by the utility company. The company gives an adjustment of $-\$0.025$ for every kilowatt-hour of electricity a customer with solar panel(s) generates that they do not use themselves.

The table shows the number of kilowatt-hours of the electricity a customer used and generated each week for a month.

	Kilowatt-Hours Used	Kilowatt-Hours Generated that were not Used by the Customer
Week 1	112.08	253.2
Week 2	181.5	357.6
Week 3	101	308
Week 4	200.42	214.4

1. What is the amount due for this month?
2. The average customer's bill is $\frac{2}{3}$ of what they would have been charged if they had not produced electricity using solar panels. Explain if this customer's savings was below or above that average.



11.7.3 Your Turn!

1. Araya is making red ribbons to sell to support a local nonprofit organization. She purchased a roll of fabric that is $30\frac{3}{4}$ feet in length and each ribbon uses 0.635 feet of fabric.

If she already has 16 ribbon orders, how many additional ribbons will she be able to sell with the single roll? Show your work.

2. Joanne purchased $2\frac{1}{4}$ pounds of tomatoes, $3\frac{1}{8}$ pounds of red onions, $\frac{3}{4}$ pounds of asparagus, and 3 lemons from a grocery store.

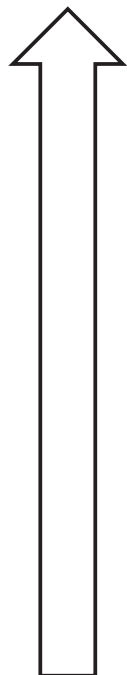
Produce	Price
Tomatoes	\$2.99 per pound
Red Onions	\$1.79 per pound
Asparagus	\$2.49 per pound
Lemons	\$0.79 each

Based on the listed prices, how much would Joanne spend?



11.7.4 Wrap-Up: Reflection

1. Color in the arrow up to the statement which best describes your current understanding of solving real-world problems with rational numbers.



I'm so confident, I could explain this to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet.

I tried hard and I listened, but I am finding this challenging. I will make sure that I get help.

I do not understand any of this yet. There are things I could do to be a better learner.



Unit 11, Lesson 8: Evaluating Numeric Expressions – Part 1



Benchmarks

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.



Learning Target

- ☐ I can evaluate numeric expressions with up to four operations.



11.8.1 Warm-Up: Math Symbols

In the 18th century, Robert Recorde decided to use a symbol for the idea of “is equal to” so he would not have to write the words over and over. As more mathematicians also began to use the symbol, the “=” became accepted as the standard symbol for equality. Math is full of symbols that shorten lengthy words and calculations. But just like symbols you use when communicating with others, like “LOL” in texting, once you learn their meaning, understanding the mathematics is not so challenging.

1. List math symbols that you are familiar with and what they represent.
2. List other math symbols you have seen before, but you are unfamiliar with.



11.8.2 Guided Instruction: Applying the Order of Operations for Expressions Including Exponents

Recall from 5th grade that an order of operations is used to evaluate numeric expressions with multiple operations.

Step 1: Perform operations within grouping symbols.

Step 2: Multiply and divide in order from left to right ($\rightarrow$).

Step 3: Add and subtract in order from left to right ($\rightarrow$).

Numerical expressions can also include exponents.

1. The expression $4 \times (-2)^3 - 8 \div (-3 + 1)$ can be rewritten by expanding the exponential expression as $4 \times (-2 \cdot -2 \cdot -2) - 8 \div (-3 + 1)$.
 - a. According to the order of operations, which expressions should be evaluated first?
 - b. When $(-2)^3$ is rewritten using multiplication, the resulting expression is in parentheses. Complete the statement.

According to the order of operations, $(-2)^3$ should

be evaluated

after
before

 multiplying by 4.

- c. Work with your partner to number each operation so that the order of operations is in the correct order.

- ☐ Perform operations inside grouping symbols.
- ☐ Perform multiplication and division in order from left to right. ($\rightarrow$)
- ☐ Evaluate expressions with exponents.
- ☐ Perform addition and subtraction in order from left to right. ($\rightarrow$)

2. Use order of operations to evaluate one of the expressions shown in the table below while your partner evaluates the other expression.

Expression A	Expression B
$(6 + 0.5)^2 \div 0.25$	$(6^2 + 0.5^2) \div 0.25$

3. How are the problems similar?

4. How are the problems different?

5. Complete the statements about the expressions.

The two expressions are ☐ equivalent
☐ not equivalent

because they ☐ have the same value.
☐ do not have the same value.

For Expression A, first

- ☐ add inside the parentheses, then apply the exponent to the sum.
- ☐ apply the exponent to each addend, then add to find the sum.

For Expression B, first

- ☐ add inside the parentheses, then apply the exponent to the sum.
- ☐ apply the exponent to each addend, then add to find the sum.

For both expressions, addition was done before division

because ☐ the addition is to the left of the division.
☐ the addition is inside the grouping symbols.

6. Anna evaluated Expression A and Expression B. Her work is shown.

Expression A	Expression B
$ -10 \div 2 + 10^2$	$-10 \div 2 + 10^2$
$10 \div 2 + 10^2$	$-10 \div 2 + 100$
$10 \div 2 + 100$	$-5 + 100$
$5 + 100$	95
105	

Anna's partner gave her a 👍 on both of her answers. Why did Anna arrive at two different values when evaluating these similar looking expressions?



Absolute value notation should be treated the same as grouping symbols when following the order of operations.

7. Zion says since multiplication is commutative and the order does not matter, he thinks he can evaluate the expression $[0.05 \times (3 \times 2)^4]^2$ by either multiplying 0.05×3 first or by multiplying 3×2 first.

a. Discuss with your partner if Zion is correct when he stated that both would work.

b. Evaluate the expression $[0.05 \times (3 \times 2)^4]^2$.



When using the order of operations to evaluate an expression with nested grouping symbols, apply the order of operations inside the inner grouping symbols first.



11.8.3 Collaboration: Determining Equivalent Expressions

Answer the questions with your partner.

Tanisha says that when she evaluated an expression, the result was 7.25.

- Use mental math to estimate each value. Then, circle which of the following expressions could have been Tanisha's expression.

$$(6 - 0.5)^2 \times 5$$

$$6 - 0.5^2 \times 5$$

$$6 + (-0.5)^2 \times 5$$

$$5 \times 6 - 0.5^2$$

- Find the values of the expressions to verify your choice.

Expression	Value
$(6 - 0.5)^2 \times 5$	
$6 - 0.5^2 \times 5$	
$6 + (-0.5)^2 \times 5$	
$5 \times 6 - 0.5^2$	



11.8.4 Exploration: Grouping Symbols

- Without working the problems, discuss with your partner which expression in the table you expect to have the largest value.

Expression A	Expression B
$10 - (0.75 \times 12) + 4^2$	$(10 - 0.75) \times (12 + 4^2)$
Expression C	Expression D
$10 - 0.75 \times 12 + 4^2$	$10 - 0.75 \times (12 + 4)^2$

- Put an asterisk (*) next to the expression you expect will have the largest value. Write your explanation here.
- Evaluate each of the expressions in the table with your partner.

4. With another set of partners, share if your guess was or was not correct. Discuss how the order of operations affected the value of your guesses. Summarize your discussion.

5. For the two expressions in the table that resulted in the same value, explain why they have the same value.



11.8.5 Your Turn!

1. Evaluate each expression without using a calculator.

a. $[(10 - 15) \times 2]^3 + 50$

b. $(2^4 - 3 \times 6)^2$

c. $|18 + (-2) \times 14|^5$

d. $[0.8 \times (1.2 + 2.8)^2]^2$

e. $[(0.8 \times 10)^2 + 4.5] \div 0.5$

f. $(10 \times 0.4)^2 + (28.4 \div (-4))$



11.8.6 Wrap-Up: Reflection

1. Complete each statement based on your learning today.

- A barrier to my learning was ...
- I will try to overcome this by ...
- Today I am still unsure about ...
- To fill in this gap I intend to ...



Unit 11, Lesson 9: Evaluating Numeric Expressions – Part 2



Benchmarks

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents, and absolute value.

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.



Learning Target

- ☐ I can evaluate numeric expressions with up to six operations.



11.9.1 Warm-Up: Bell Work

Explain the steps needed to evaluate the expressions.

1. $(10 + 18 \cdot 2)^5 \div 3$

2. $[(7 - 10)^2]^4 \times 2$



11.9.2 Guided Instruction: Evaluating Expressions with Multiple Steps

Two students evaluated the expression below in two different ways. Review the work of each student. Then, answer the questions.

Diego	Valeria
$[0.5 \times (0.5 + 3.5)^2]^3 + 26 \div 13$	$[0.5 \times (0.5 + 3.5)^2]^3 + 26 \div 13$
$[0.5 \times 4^2]^3 + 26 \div 13$	$0.5^3 \times (0.5 + 3.5)^{2 \times 3} + 26 \div 13$
$[0.5 \times 16]^3 + 26 \div 13$	$0.5^3 \times (0.5 + 3.5)^6 + 26 \div 13$
$8^3 + 26 \div 13$	$0.5^3 \times 4^6 + 26 \div 13$
$512 + 26 \div 13$	$0.125 \times 4^6 + 26 \div 13$
$512 + 2$	$0.125 \times 4,096 + 26 \div 13$
514	$512 + 26 \div 13$
	$512 + 2$
	514

- How were Diego's steps and Valeria's steps different from each other?
- Discuss with your partner whether you think one method was easier to use to evaluate the expression over the other.



11.9.3 Activity: Expression Game

- Each group will use a standard number cube with sides numbered 1 – 6 to play a game. One player rolls the number cube. Each player decides which box to place that number in the expression. Continue rolling the cube until all blanks in the expression are filled. Evaluate your expression and record your score based on the point system in the table. If on any round two players' expressions have the same value, both players **lose 10 points**.

Round 1 Scoring – the player with the greatest value earns 8 points.	$[(_\times_\^2)+_\]\div_\]$	My answer _____ Points earned _____
Round 2 Scoring – the player with the least value earns 5.5 points.	$[(_\times_\)^2+_\]\div_\]$	My answer _____ Points earned _____
Round 3 Scoring – the player with the greatest absolute value loses 3.7 points.	$(_\times_\^2)-(_\div_\)$	My answer _____ Points earned _____
Round 4 Scoring – the player with the absolute value closest to zero earns 11 points.	$(_\times_\)^2-(_\div_\)$	My answer _____ Points earned _____

My Game Score

Round 1	Round 2	Round 3	Round 4	My Score

- Discuss with the other players what strategies they used to determine where to put the numbers. Summarize your discussion.



11.9.4 Try It: Evaluating Expressions with Multiple Steps

1. In the table an expression has been evaluated, but the steps are mixed up. Order the expressions using the numbers 1 – 7 on the left. Write an explanation for each step.

Step	Expression	Explain the Step
	$ -11.591 + 15 - (-3.1)^2$	
	$11.591 + 15 - 9.61$	
	$11.591 + 15 - (-3.1)^2$	
1	$ 18.2 + (-3.1)^3 + 15 - (-3.1)^2$	Original expression
	$26.591 - 9.61$	
	$ 18.2 + (-29.791) + 15 - (-3.1)^2$	
	16.981	

2. Evaluate each expression without using a calculator.

a. $\left(\frac{3}{4}\right)^2 \times 32 + 3^2$

b. $[(10 + 4.5)^2 - 10.25]^2 \div 40$

c. $6.7 - (2^3 - 3)^3 \div 10 + 4$



11.9.5 Activity: Class Debate

James wants to choose an expression that has a value of -778 . He can choose between the expressions shown.

Expression A	Expression B
$14 - (10 \times 3^2)^2 + 6 \times 3$	$14 - 10 \times (3^2)^2 + 6 \times 3$
Expression C	Expression D
$(14 - 10) \times (3^2 + 6)^2 \times 3$	$14 - [10 \times (3^2)^2] + 6 \times 3$

Which expression should James choose? _____

- Each corner of the room is labeled A, B, C, or D. After selecting the expression that you think James should choose, go to the corner of the room that matches the label for that expression.
- With your classmates who selected the same expression, discuss why your expression is the expression that James should choose. Discuss why James should not choose the other expressions. One member of your group will act as a representative during a class debate.
- For the debate, prepare an opening statement and potential rebuttals for arguments from other groups. Keep arguments and rebuttals centered on the mathematics.
- After the debate, explain whether you would select the same expression again.



11.9.6 Wrap-Up: Ticket Out the Door

1. Evaluate each expression without a calculator. Show each step you use.

$$-\left|5 - \frac{3}{5}\right| + \frac{4}{15}(4^2 - 61)$$

Work

$$\frac{(1.5 + |-3.25 - 4.25|)^3}{12}$$

Work

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